EE-708 : QUIZ-1

WEDNESDAY

APR '23

12

102-263 WK-15

1) i) 'X' takes 8 values.

For a fixed length code, which is binary and error free,

Each symbol is represented by $\lceil \log_2 M \rceil$, (1M)

where $M = 8$ distinct values here.

$\therefore$

Min. avg. length = 3

ii) We note that Huffman code will allow us to find the min. avg. length in this situation.

(0.5 M)

Since we care only about length, we can omit the codeword for each alphabet symbol.

* Look at the number of combinations in prob. that symbol is involved in.

Length	X	Prob.						
2	6	0.31	0.31	0.31	0.31	0.31	0.42	0.58 → 1
2	1	0.22	0.22	0.22	0.22	0.27	0.31	0.42
3	3	0.12	0.12	0.15	0.2	0.22	0.27	
3	5	0.1	0.1	0.12	0.15	0.2		
3	7	0.1	0.1	0.1	0.12			
4	2	0.08	0.08	0.1				
5	4	0.04	0.07					
5	8	0.03						

(1M for construction)

Avg. length $= 2\times0.31 + 2\times0.22 + 3\times0.12 + 3\times0.1 + 3\times0.1$
$+ 4\times0.08 + 5\times0.04 + 5\times0.03$
$= 2.69$ bits (0.5M)

Though $H(X) = 2.66$, that limit is not achievable for symbol-wise coding

iii) For a ternary code, we need the total symbols to be $2k+1$, $k \in \mathbb{Z}^+ \cup \{0\}$
Add $X = 9$ as a zero probability symbol.
(0.5M)

Note: Constructions in tree-form are acceptable, as long as the probability at each node is written correctly.

X	Prob.					Codeword
6	0.31	0.31	0.31	0.44	1	1
1	0.22	0.22	0.25	0.31		0 0
3	0.12	0.12	0.22	0.25		0 1
5	0.1	0.1	0.12			0 2
7	0.1	0.1	0.1			2 0
2	0.08	0.08				2 1
4	0.04	0.07				2 2 0
8	0.03					2 2 1
9	0					2 2 2

TERNARY HUFFMAN CODE

Q2. Codebook: {0, 01}

a) No

'0' is a prefix of '01', hence they are not instantaneously decodable.

b) Yes.

Given a complete received string, the decoder can replace all '01' observations as '1'. The remaining '0' stay as is. This final string is the uniquely ~~decodable~~ decoded output.

c) Yes.

Since the codebook has distinct codewords.

Solutions to EE 708 Quiz I: Questions 3 & 4

Question 3

Problem: Let X and Y be two random variables having a joint probability mass function as shown below. Compute $H(X)$, $H(Y)$, $H(X,Y)$, $I(X;Y)$, $H(X|Y)$, and $H(Y|X)$.

$X \setminus Y$	**0**	**1**	**2**
0	0	1/4	1/4
1	1/4	1/4	0

Step 1: Calculate Marginal Probabilities

First, we find the marginal distributions $P(X)$ by summing the rows and $P(Y)$ by summing the columns.
Marginal of X:

$$P(X=0) = 0 + \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$
$$P(X=1) = \frac{1}{4} + \frac{1}{4} + 0 = \frac{1}{2}$$

Thus, $X \sim \{\frac{1}{2}, \frac{1}{2}\}$.
Marginal of Y:

$$P(Y=0) = 0 + \frac{1}{4} = \frac{1}{4}$$
$$P(Y=1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$
$$P(Y=2) = \frac{1}{4} + 0 = \frac{1}{4}$$

Thus, $Y \sim \{\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\}$.

Step 2: Compute Entropies

1. Entropy of X, $H(X)$ (0.5 marks):

$$H(X) = -\sum_x p(x)\log_2 p(x) = -\left(\frac{1}{2}\log_2\frac{1}{2} + \frac{1}{2}\log_2\frac{1}{2}\right) = 1 \text{ bit}$$

2. Entropy of Y, $H(Y)$ (0.5 marks):

$$\begin{aligned} H(Y) &= -\sum_y p(y) \log_2 p(y) \\ &= -\left(\frac{1}{4}\log_2\frac{1}{4} + \frac{1}{2}\log_2\frac{1}{2} + \frac{1}{4}\log_2\frac{1}{4}\right) \\ &= -\left(\frac{1}{4}(-2) + \frac{1}{2}(-1) + \frac{1}{4}(-2)\right) \\ &= -(-0.5 - 0.5 - 0.5) = 1.5 \text{ bits} \end{aligned}$$

3. Joint Entropy $H(X,Y)$ (0.5 marks): Summing over the non-zero entries in the joint table (four entries, each equal to 1/4):

$$\begin{aligned} H(X,Y) &= -\sum_{x,y} p(x,y) \log_2 p(x,y) \\ &= -4 \times \left(\frac{1}{4}\log_2\frac{1}{4}\right) \\ &= -\log_2(2^{-2}) = 2 \text{ bits} \end{aligned}$$

Step 3: Mutual Information and Conditional Entropies

4. Mutual Information $I(X;Y)$ (0.5 marks):

$$I(X;Y) = H(X) + H(Y) - H(X,Y) = 1 + 1.5 - 2 = 0.5 \text{ bits}$$

5. Conditional Entropy $H(X|Y)$ (0.5 marks):

$$H(X|Y) = H(X,Y) - H(Y) = 2 - 1.5 = 0.5 \text{ bits}$$

6. Conditional Entropy $H(Y|X)$ (0.5 marks):

$$H(Y|X) = H(X,Y) - H(X) = 2 - 1 = 1 \text{ bit}$$

Question 4

Problem: Let $q_{\mathbf{x}}$ denote the empirical distribution of the vector $\mathbf{x}$. Prove that the maximum likelihood (ML) condition:

$$Pr\{\mathbf{X} = \mathbf{x}|\theta = 0\} > Pr\{\mathbf{X} = \mathbf{x}|\theta = 1\} \tag{1}$$

is equivalent to the condition $D(q_{\mathbf{x}}||p_0) < D(q_{\mathbf{x}}||p_1)$.

Proof

We start directly with the Maximum Likelihood inequality:

$$\prod_{i=1}^{n} p_0(x_i) > \prod_{i=1}^{n} p_1(x_i)$$

Taking the logarithm (base 2) on both sides:

$$\sum_{i=1}^{n} \log p_0(x_i) > \sum_{i=1}^{n} \log p_1(x_i)$$

We can rewrite the sum over the sequence indices $i = 1 \dots n$ as a sum over the alphabet $\mathcal{X}$, weighted by the count of each symbol $N(a|\mathbf{x})$. Substituting $N(a|\mathbf{x}) = nq_{\mathbf{x}}(a)$:

$$\sum_{a \in \mathcal{X}} nq_{\mathbf{x}}(a) \log p_0(a) > \sum_{a \in \mathcal{X}} nq_{\mathbf{x}}(a) \log p_1(a)$$

Dividing by n:

$$\sum_{a \in \mathcal{X}} q_{\mathbf{x}}(a) \log p_0(a) > \sum_{a \in \mathcal{X}} q_{\mathbf{x}}(a) \log p_1(a) \quad \textbf{(1 marks)}$$

To relate this to KL divergence, we subtract the entropy term $\sum_{a \in \mathcal{X}} q_{\mathbf{x}}(a) \log q_{\mathbf{x}}(a)$ from both sides.

$$\sum_{a} q_{\mathbf{x}}(a) \log p_0(a) - \sum_{a} q_{\mathbf{x}}(a) \log q_{\mathbf{x}}(a) > \sum_{a} q_{\mathbf{x}}(a) \log p_1(a) - \sum_{a} q_{\mathbf{x}}(a) \log q_{\mathbf{x}}(a)$$

Factor out the negative sign on both sides:

$$-\left(\sum_{a} q_{\mathbf{x}}(a) \log q_{\mathbf{x}}(a) - \sum_{a} q_{\mathbf{x}}(a) \log p_0(a)\right) > -\left(\sum_{a} q_{\mathbf{x}}(a) \log q_{\mathbf{x}}(a) - \sum_{a} q_{\mathbf{x}}(a) \log p_1(a)\right)$$

Using the definition of KL divergence $D(q||p) = \sum q(x) \log \frac{q(x)}{p(x)}$, this becomes:

$$-D(q_{\mathbf{x}}||p_0) > -D(q_{\mathbf{x}}||p_1)$$

Multiplying by -1 reverses the inequality:

$$D(q_{\mathbf{x}}||p_0) < D(q_{\mathbf{x}}||p_1) \quad \textbf{(1 marks)}$$

Thus, the condition is proven. $\square$

Approach from either LHS to RHS is valid and will fetch 1 mark to reach the mid-point and full credit to reach the end point. Partial credit will be awarded as per the progress.